

Mock Midterm B

Name: _____

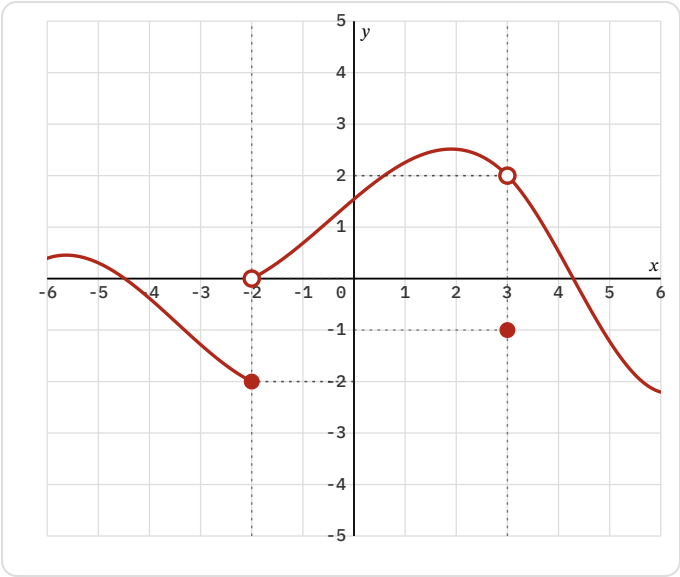
Date: _____

Score: _____ / 58

Directions: Answer each question below. Be sure to show work to receive full credit. Work this exam on your own, without notes or AI, the way you will on the proctored midterm. Be sure the notation in your solutions matches the notation from the lesson videos. Write **DNE** if a value or limit does not exist. Suggested time: 75 minutes.

1.

Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write **DNE**.



a.) $\lim_{x \rightarrow -2^-} f(x)$ _____ (2)

b.) $\lim_{x \rightarrow 3^+} f(x)$ _____ (2)

4 pts

2. Let $f(x) = \frac{x^3-27}{x-3}$. Use the table to see what the outputs do as x approaches **3**. 3 pts

x	2.99	2.999	2.9999	3	3.0001	3.001	3.01
$f(x)$	26.9101	26.991001	26.9991	?	27.0009	27.009001	27.0901

a.) Can you find $\lim_{x \rightarrow 3} f(x)$ just by plugging in $x = 3$? (1)

(A) No — $x = 3$ makes a denominator 0, so $f(3)$ is undefined

(B) Yes — $f(3)$ is defined, so substitute

b.) $\lim_{x \rightarrow 3} f(x) =$ _____ (2)

3. Calculate each limit algebraically. 4 pts

a.) $\lim_{x \rightarrow -4} (2x^2 + 3x + 2)$ (2)

b.) $\lim_{x \rightarrow 2} \sqrt{x^2 + 21}$ (2)

4.

Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE.

a.) $\lim_{x \rightarrow 16} \frac{x-16}{\sqrt{x}-4}$ _____ (2)

b.) $\lim_{h \rightarrow 0} \frac{(2+h)^3-8}{h}$ _____ (2)

4 pts

5.

Compute each limit algebraically. Use ∞ / $-\infty$ for unbounded results.

4 pts

a.) $\lim_{x \rightarrow -\infty} \frac{4x^2+1}{6x^2-9}$ _____ (2)

b.) $\lim_{x \rightarrow \infty} \frac{x^5-7x^4+7x^3-3x^2+7x+4}{-4x^3+9x^2}$ _____ (2)

6.

Let

4 pts

$$G(x) = \begin{cases} -x^2 & x < -1 \\ 2x + 1 & x > -1 \end{cases}$$

Find each quantity. If it does not exist, write **DNE**.

a.) $G(-1)$ _____ (2)

b.) $\lim_{x \rightarrow \infty} G(x)$ _____ (2)

7.

Let

4 pts

$$f(x) = \begin{cases} \frac{x^2 - 2x - 8}{x + 2} & x < -2 \\ 3x + 2 & x \geq -2 \end{cases}$$

Find each limit or function value algebraically. Write **DNE** if it doesn't exist.

a.) $\lim_{x \rightarrow 0} f(x)$ _____ (2)

b.) $\lim_{x \rightarrow -2} f(x)$ _____ (2)

8.

Find the value of k that makes f continuous at $x = -2$.

4 pts

$$f(x) = \begin{cases} x^2 + x & x < -2 \\ kx + 6 & x \geq -2 \end{cases}$$

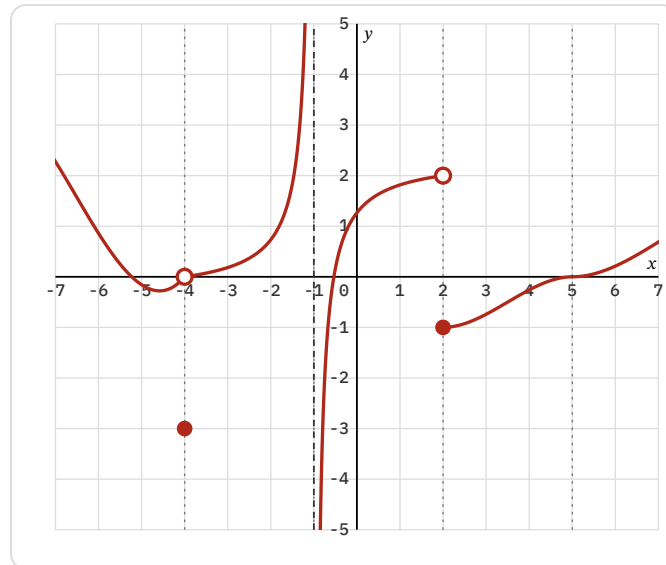
a.) $\lim_{x \rightarrow -2^-} f(x)$ _____ (2)

b.) $k =$ _____ (2)

9.

Classify the behavior of f at each labeled x -value.

2 pts



a.) At $x = -4$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

b.) At $x = 5$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

10.

Let $V(t) = t^2 - 50t + 300$, where $V(t)$ represents the number of gallons of water in a tank t minutes after a drain is opened. Find the average rate of change over $[6, 8]$.

a.) $V(6)$ _____ (2)

b.) Which sentence correctly states the result **with units**? (1)

(A) The amount of water in the tank changes by a total of -36 gallons from $t = 6$ to $t = 8$.

(B) At exactly $t = 8$ minutes, the amount of water in the tank is changing at a rate of -36 gallons per minute.

(C) Between $t = 6$ and $t = 8$ minutes, the amount of water in the tank changes at an average rate of -36 gallons per minute.

(D) Between $t = 6$ and $t = 8$, the amount of water in the tank changes at an average rate of -36 minutes per gallon.

3 pts

11.

Let $f(x) = \frac{-2}{x}$. Calculate the average rate of change of f over each interval. (Rounded answers to 2+ decimal places are fine.)

a.) $[3, 5]$ _____ (2)

b.) As the interval shrinks toward $x = 3$, these values approach... (1)

(A) $\frac{2}{9}$

(B) $-\frac{2}{9}$

(C) $\frac{4}{9}$

(D) $\frac{11}{9}$

3 pts

12.

Let $f(x) = x^3 - 3x$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it to fill in the table.

a.) Simplified difference quotient (in terms of x and h) _____ (3)

b.) As $h \rightarrow 0$, what value do these approach? _____ (2)

5 pts

13.

Let $f(x) = 3x^2 - 2x - 6$. Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$.

a.) $f'(x) =$ _____ (3)

b.) What does $f'(-3) = -20$ mean? (1)

(A) The slope of the secant line from $x = -3$ to $x = -3 + 1$ is -20 .

(B) The value of f at $x = -3$ is -20 .

(C) The average rate of change of f from $x = 0$ to $x = -3$ is -20 .

(D) The instantaneous rate of change of f at $x = -3$ is -20 — the slope of the tangent line there.

4 pts

14. Let $f(x) = x^2 - 8x - 3$, so $f'(x) = 2x - 8$. 4 pts

a.) At what x is the tangent line horizontal? (2)

b.) At what x does the tangent line have slope 2? (2)

15.

Let $h(t) = -16t^2 + 128t$ be the height, in feet, of a rocket t seconds after launch.

a.) $h'(t) =$ _____ (3)

b.) Interpret $h'(5) = -32$. (1)

(A) Over the first 5 seconds, the height of the rocket changed at an average rate of 32 feet per second.

(B) At $t = 5$ seconds, the height of the rocket is 32 feet.

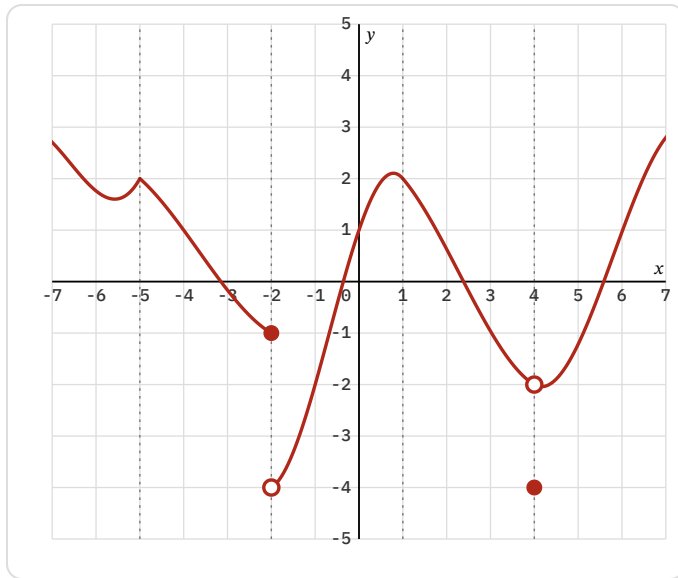
(C) At $t = 5$ seconds, the height of the rocket is decreasing at a rate of 32 feet per second.

(D) At $t = 5$ seconds, the height of the rocket is increasing at a rate of 32 feet per second.

4 pts

16.

At each labeled x -value, decide whether the graph of f is continuous there and whether it is differentiable there.



a.) At $x = -5$

(1)

- (A) Continuous and differentiable
- (B) Continuous but NOT differentiable
- (C) Not continuous (so not differentiable)

b.) At $x = 4$

(1)

- (A) Continuous and differentiable
- (B) Continuous but NOT differentiable
- (C) Not continuous (so not differentiable)

2 pts
